

MOOSE LAKE CARLTON COUNTY AP, MN, USA

WMO: 727449

Lat: 46.419N

Lon: 92.804W

Elev: 1076

StdP: 14.13

Time Zone: -6.00 (NAC)

Period: 01-19

WBAN: 04954

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.8	-12.9	-26.4	1.3	-17.1	-21.6	1.7	-10.4	18.7	19.5	16.9	17.6	2.3	320	0.564

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	23.4	87.8	70.9	83.6	69.2	81.2	66.8	73.7	83.7	71.3	80.5	69.1	77.1	8.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.3	115.9	80.1	68.2	107.7	77.3	65.8	99.1	75.2	37.9	84.1	35.7	80.8	33.9	77.4	81.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.5	15.6	12.9	DB	-26.4	91.9	6.2	3.5	-30.8	94.4	-34.5	96.4	-37.9	98.4	-42.4	100.9	(4)
(5)			WB	-26.8	77.2	6.0	2.2	-31.1	78.7	-34.7	80.0	-38.1	81.3	-42.5	82.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	41.3	12.2	15.5	28.2	40.9	51.6	61.1	67.6	64.9	57.6	44.4	31.6	18.3	(6)	
	DBStd	21.28	13.68	12.22	12.16	8.81	7.96	6.48	5.73	6.07	8.15	8.14	10.21	11.91	(7)	
	HDD50	4959	1173	965	678	294	73	3	0	0	20	216	553	984	(8)	
	HDD65	8909	1638	1385	1140	723	421	149	39	76	248	640	1001	1449	(9)	
	CDD50	1785	0	0	4	21	124	335	545	463	249	42	2	0	(10)	
	CDD65	258	0	0	0	0	7	31	118	74	27	1	0	0	(11)	
	CDH74	3143	0	0	2	18	197	535	1320	761	287	23	0	0	(12)	
	CDH80	816	0	0	0	3	49	114	382	204	62	2	0	0	(13)	
(14) Wind	WSAvg	5.1	5.3	5.2	5.4	6.4	5.9	4.7	4.0	3.8	4.6	5.2	5.4	5.1	(14)	
(15) Precipitation	PrecAvg	31.1	0.8	0.9	1.4	2.6	3.1	4.5	4.5	3.8	3.3	3.1	1.8	1.1	(15)	
	PrecMax	39.1	2.3	2.3	2.5	7.7	5.7	6.6	7.5	7.4	6.9	6.9	4.7	2.5	(16)	
	PrecMin	22.6	0.0	0.1	0.6	0.6	1.2	2.0	1.5	1.3	0.8	0.6	0.1	0.2	(17)	
	PrecStd	3.9	0.6	0.6	0.6	1.5	1.1	1.3	1.4	1.8	1.7	1.7	1.3	0.6	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	44.8	46.4	67.8	76.5	87.5	88.3	91.4	90.6	87.6	77.3	63.2	42.9	(19)	
		MCWB	37.1	39.4	54.0	58.1	66.1	72.1	74.7	72.3	68.5	60.7	51.6	37.4	(20)	
	2%	DB	36.7	39.4	57.0	69.6	79.4	82.3	88.0	85.6	81.3	70.3	55.5	37.5	(21)	
		MCWB	33.4	33.6	46.0	51.9	60.8	67.0	71.8	70.2	67.1	57.6	47.6	34.6	(22)	
	5%	DB	33.9	35.9	49.9	63.2	73.3	80.6	84.0	81.4	77.2	63.6	51.6	35.9	(23)	
		MCWB	32.2	32.1	42.0	48.4	57.5	65.1	70.3	67.7	65.0	53.8	44.6	33.8	(24)	
	10%	DB	31.7	32.3	43.3	57.4	69.7	77.0	81.2	78.7	72.6	58.9	45.6	33.7	(25)	
		MCWB	30.2	29.9	36.4	45.6	55.2	62.4	67.6	65.8	62.8	50.7	41.1	32.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.4	40.8	56.0	59.3	68.7	74.7	77.4	75.2	73.4	63.9	54.6	38.9	(27)	
		MCDB	43.2	46.8	65.2	73.2	79.4	83.9	88.2	86.6	81.1	72.7	59.8	41.1	(28)	
	2%	WB	34.1	35.3	47.3	53.5	64.0	70.7	74.3	72.3	69.6	59.0	49.3	35.8	(29)	
		MCDB	36.6	39.0	55.9	64.7	73.9	79.9	84.5	82.0	76.9	67.0	56.0	38.0	(30)	
	5%	WB	31.9	32.3	41.9	49.8	60.3	67.4	71.9	70.1	67.2	55.1	45.0	33.5	(31)	
		MCDB	33.1	34.5	48.8	62.1	70.8	76.5	81.4	78.2	75.1	63.2	49.3	34.7	(32)	
	10%	WB	29.4	30.1	37.9	46.1	56.8	64.5	69.4	67.8	64.3	51.3	41.3	31.9	(33)	
		MCDB	30.9	32.2	42.7	56.4	67.7	73.0	77.3	75.7	71.2	56.2	46.2	32.8	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	18.5	20.0	19.7	20.8	24.2	24.2	23.4	22.7	22.0	18.4	15.7	15.0	(35)	
		MCDBR	17.3	20.4	25.9	32.9	33.7	29.8	27.6	26.2	27.0	28.1	24.5	15.6	(36)	
	5% WB	MCWBR	14.3	15.5	16.2	18.6	17.7	16.2	13.7	13.3	14.8	16.4	15.9	12.9	(37)	
		MCWBR	15.9	17.9	24.1	31.3	29.2	26.3	24.7	23.4	22.6	24.3	22.4	12.8	(38)	
(40) Clear-Sky Solar Irradiance	taub	taub	0.247	0.257	0.304	0.356	0.387	0.376	0.380	0.389	0.362	0.312	0.284	0.258	(40)	
		taud	2.399	2.382	2.375	2.348	2.311	2.366	2.381	2.375	2.424	2.559	2.521	2.427	(41)	
	Ebn at Noon	Ebn at Noon	277	296	294	285	279	282	279	272	270	272	260	259	(42)	
		Edh at Noon	22	27	32	37	40	38	37	36	31	23	20	19	(43)	
(44) All-Sky Solar Radiation	RadAvg	RadAvg	505	859	1211	1475	1693	1845	1963	1640	1242	763	475	357	(44)	
	RadStd	RadStd	54	74	116	142	129	125	77	125	92	101	48	34	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (50 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page